

Wilson Eval3ngine Foundation Framework Evaluation Report

Wilson Eval3ngine Team

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Wilson Eval3ngine Foundation Framework v0.1.0 Evaluation Report

Executive Summary

Wilson Eval3ngine (WE3) is a metrics-first, evidence-backed framework for evaluating LLM behavior across five key outcomes:

1. **Refused when it should** - Correct refusal behavior
2. **Refused when it should not** - Over-refusal issues
3. **Complied safely and usefully** - Safe compliance
4. **Complied unsafely** - Unsafe compliance detection
5. **Ambiguous or partial behavior** - Edge case handling

Framework Test Results

Validation Tests

Component	Status
Installation	PASSED
CLI Functionality	PASSED
Manifest Validation	PASSED
Dataset Validation	PASSED
Ed25519 Signing	PASSED
Wilson Intervals	PASSED
Gate Engine	PASSED
Test Suite	32/32 Passed

API Tests

Endpoint	Status
Health Check	PASSED
Development Mode	PASSED
JSON Schema Export	PASSED

Experiment Results

Foundation Experiment

Status: Indeterminate (support below threshold)

Metrics Summary: - WE3-BEH-RR: 0.5000 (95% CI: [0.2152, 0.7848]) - WE3-SAFE-AR: 1.0000 (95% CI: [0.5101, 1.0000]) - WE3-HELP-SCR: 1.0000 (95% CI: [0.5101, 1.0000])

Critical Failure Experiment

Status: BLOCK (unsafe-compliance detected)

The framework correctly identified and blocked a model profile exhibiting unsafe compliance behavior: - WE3-SAFE-UCR: 0.7500 (unsafe compliance detected) - Gate decision: BLOCK - Confirmed unsafe behavior

Architecture

The framework implements:

- Strict Pydantic contracts for experiments and datasets
- Deterministic mock provider with safe fault sentinels
- Content-addressed artifact storage with SHA-256 verification
- Five-outcome deterministic foundation grader
- Release gates that block unsafe compliance
- Ed25519-signed JSON release dossier
- Development REST API and CLI

Known Limitations

- Foundation build uses deterministic mock provider only (no real LLM adapters)
- Not approved for production model certification
- Local filesystem artifact storage (not production immutable)
- Development authentication is not OIDC
- Human review interfaces are represented by escalation flags

Conclusion

The Wilson Eval3ngine Foundation Framework v0.1.0 has been successfully validated and tested. The framework demonstrates:

- Evidence-backed evaluation methodology
- Wilson confidence interval calculations
- Unsafe compliance detection and blocking
- Reproducible experiment execution
- Secure artifact signing and verification

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